

grid-meter-app — Deployment Topology

Where each resource actually runs, as of 2026-09-18 — laptop (self-hosted, both tracks) vs. AWS (Terraform-provisioned public cloud)

LAPTOP (local, self-hosted)

Two separate local tracks — same hardware, different purposes (hands-on HA re-familiarization vs. k8s interview demo)

Docker Compose

day-to-day dev

▶ Traefik — edge routing + load balancing

▶ React SPA (served by Nginx)

▶ API — Spring Boot, 2x Tomcat replicas

▶ Kafka — multi-broker, KRaft mode

▶ Redis + Sentinel — self-built HA

▶ PostgreSQL + Patroni + Consul — self-built HA

▶ Prometheus, Grafana, Loki, Tempo, Alloy — full observability stack

kind cluster

k8s interview demo

▶ Traefik (Ingress controller)

▶ API, Frontend, Kafka, Postgres, Redis — all in-cluster

▶ kube-prometheus-stack (Helm) + Loki/Tempo/Alloy — in-cluster

JMeter runs natively on the laptop host (not containerized) — generates load against either track above.

separate,
independent
tracks

AWS — us-west-2 (public cloud)

Real, billed infrastructure — Terraform-provisioned, independently reachable, spun up/torn down per interview session

Terraform-managed

infrastructure

▶ VPC — 3 AZs, public+private subnets, 1 NAT gateway

▶ EKS cluster + managed node group (3x t3.medium)

▶ RDS PostgreSQL (managed — replaces self-hosted Patroni)

▶ ElastiCache Valkey (managed — replaces self-hosted Sentinel)

▶ ECR — api/frontend image repositories

▶ EBS CSI driver + gp3/XFS StorageClass

kubectl-deployed

app layer (in-cluster)

▶ Traefik — fronted by a real Classic ELB (public entry point)

▶ API (2 replicas), Frontend

▶ Kafka — self-hosted, 3-broker StatefulSet, EBS-backed (deliberately not managed — see cloud-deployment-scope.md)

⛔ Deliberately NOT deployed here

× Observability (Prometheus/Grafana/Loki/Tempo/Alloy) — kind-only, by choice

× Load testing (JMeter) — never run against this deployment

Laptop — solid border, self-hosted, always available AWS — dashed border, real billed cloud infra, spun up/torn down per session Explicitly out of scope for that target (not a gap — a documented decision) **managed** = cloud provider's managed service, replacing a self-hosted equivalent